Microsoft Configuration Manager (SCCM)

Microsoft Configuration Manager, widely known by its former name System Center Configuration Manager (SCCM), is Microsoft's enterprise platform for managing large numbers of Windows devices and servers from one central console. Over the years the product has been called SMS, SCCM, Microsoft Endpoint Configuration Manager and now Microsoft Configuration Manager, but in the industry (and in this book) the name SCCM is still the most common. Today it is delivered as the "Current Branch", which receives regular version updates directly inside the console, and it is part of the Microsoft Intune family of products.

In a small network an administrator can walk to each computer to install software or apply updates. In an enterprise with hundreds or thousands of devices spread across many locations, this is impossible. SCCM solves this problem by letting the IT team do the following from a single place:

- **Inventory:** collect hardware and software details from every managed device.
- **Software distribution:** install, update and remove applications and packages.
- **Software updates:** approve, deploy and track Windows and Microsoft product patches (it builds on the WSUS role covered earlier in this book).
- **Operating system deployment:** build and deploy Windows images using task sequences (an enterprise alternative to WDS).
- **Compliance settings:** define configuration baselines and report which devices do not follow them.
- **Endpoint protection management:** manage Microsoft Defender settings and policies.
- **Remote tools and reporting:** remote control, Resource Explorer and detailed reports about the whole environment.

SCCM is tightly integrated with Active Directory, DNS, SQL Server, IIS and WSUS, so it is normally deployed after those services are stable. Understanding how these pieces fit together is the foundation for everything else in this chapter.

SCCM Architecture

An SCCM environment is built from a small number of core components. In a lab or a small company all of them can run on a single server, while in large organizations they are spread across many servers and sites.

Component	Purpose
Site Server	The main server where SCCM is installed. It runs the core services, processes the data that arrives from clients and manages all other site systems.
Site Database	A Microsoft SQL Server database (named CM_<SiteCode>) that stores all configuration data, inventory, deployments and status information. It is the "brain" of the site.

Component	Purpose
SMS Provider	A WMI interface that lets the console (and scripts) read and write data in the site database. The console never talks to the database directly.
Configuration Manager Console	The administrator's tool, installed on the site server or on an admin workstation. All tasks are performed here.
Management Point (MP)	The communication point between clients and the site. Clients download policies from the MP and send inventory and status messages back to it.
Distribution Point (DP)	A file server that stores the content (applications, packages, boot images, updates) which clients download.
Client	The SCCM agent installed on every managed computer. It runs as the SMS Agent Host service (CcmExec).

How the components work together

- **Policy:** the administrator creates a deployment in the console. The site stores it in the database, and clients periodically contact the Management Point to download their policy.
- **Content:** when the client sees it must install something, it asks the Management Point which Distribution Point is closest (decided by boundary groups) and downloads the files from there.
- **Inventory and status:** the client sends hardware inventory, software inventory and status messages to the Management Point. They pass through inbox folders on the site server and are finally written to the SQL database, where the console and reports can read them.

Installation Prerequisites

Most SCCM installation failures are caused by missing prerequisites, so planning this stage carefully saves a lot of time. The exact supported versions change with each release, therefore always confirm them in Microsoft's "Supported configurations" documentation for the version being installed. The table below lists the typical requirements for a stand-alone primary site.

Category	Requirement
Server and OS	A dedicated Windows Server (2016 or later, check the supported list) with a static IP address, joined to the domain. The server name cannot be changed after installation.
Hardware (lab)	About 4 vCPU, 16 GB RAM and 150 GB or more of disk space. Production servers need much more, with separate disks for the OS, SQL data/logs and the content library.

Category	Requirement
Active Directory	Domain functional level supported by the version in use. The AD schema can be extended with `extadsch.exe` (optional but recommended) and a System Management container should be created in the domain, giving the site server computer account Full Control on it.
SQL Server	A supported SQL Server version (for example 2019 or 2022) with collation `SQL_Latin1_General_CP1_CI_AS`, Database Engine installed, and Windows Authentication enabled. Limit SQL maximum memory (Microsoft recommends giving SQL 50–80% of the server's RAM).
Windows features	IIS (with ISAPI extensions, Windows Authentication and IIS 6 Metabase/WMI compatibility), BITS, Remote Differential Compression, and .NET Framework 4.8 (minimum 4.6.2).
Windows ADK	The Windows Assessment and Deployment Kit and the WinPE add-on matching the SCCM version. They are required for operating system deployment and boot images.
Accounts	The installing account must be a local administrator on the site server and have the sysadmin role in SQL Server.
Firewall and ports	TCP 445 (SMB), 1433 (SQL), 4022 (SQL Service Broker), 80/443 (HTTP/HTTPS to MP and DP) and 10123 (client notification) must be open between the relevant servers and clients.

Installation Process

Once all prerequisites are ready, SCCM is installed using a setup wizard. The installation media can be downloaded from the Microsoft Evaluation Center (180-day evaluation) or the Volume Licensing portal for licensed use. The main steps are:

1. Run `splash.hta` from the media and choose **Install**. Use the **Download required prerequisite files** option first and save them to a local folder so the installer does not need internet access later.
2. On the **Available Setup Options** page select **Install a Configuration Manager primary site**.
3. Enter the product key, or choose the evaluation edition, and accept the license terms.
4. Provide the path to the downloaded prerequisite files and select the server and client languages.
5. Set the **Site Code** (three characters, for example `PS1`), **Site Name** and the installation folder. Make sure **Install the Configuration Manager console** is ticked.
6. Select **Install the primary site as a stand-alone site** (a hierarchy is not needed for most organizations).
7. Enter the SQL Server name (FQDN), instance name and database name (default `CM_<SiteCode>`). The Service Broker port is 4022 by default.
8. Confirm the SMS Provider location, normally the site server itself.

9. Choose the client communication method. Recent versions require **HTTPS or Enhanced HTTP**, so select the option that matches the environment's certificate setup.
10. Install the Management Point and Distribution Point on the site server (acceptable in a lab and small sites).
11. Review the summary and run the **Prerequisite Check**. Fix every error and review each warning, then click **Begin Install**.

The installation normally takes 30–60 minutes. Its progress can be followed in the log file `C:\ConfigMgrSetup.log` and the prerequisite check results in `C:\ConfigMgrPrereq.log`. When it finishes, open the console and confirm that the site shows as active under **Monitoring > Overview > System Status > Site Status**.

Post Installation Tasks

A freshly installed site manages nothing yet. The following tasks turn it into a working environment:

- **Install updates and hotfixes:** open **Administration > Updates and Servicing** and install the latest available version and hotfixes.
- **Configure discovery methods:** enable Active Directory System Discovery, User Discovery and Group Discovery (plus Heartbeat Discovery) so SCCM learns about the computers and users in the domain.
- **Create boundaries and boundary groups:** boundaries (AD site, IP subnet or IP range) tell clients which site systems to use. Without them, content downloads and site assignment fail.
- **Add site system roles:** install additional Distribution Points, the Software Update Point (linked to WSUS) and the Reporting Services Point as needed.
- **Configure accounts:** create the Client Push Installation Account and other service accounts using the least privilege principle.
- **Customize client settings:** adjust the default client settings (policy polling, inventory schedules, software update behavior) and create custom settings for specific collections.
- **Enable maintenance tasks:** most importantly the **Backup Site Server** task, which creates a regular backup of the site database and configuration (see the Backup & Recovery chapter).

SCCM Hierarchy and Roles

A hierarchy describes how multiple SCCM sites are arranged. There are three types of sites:

Site type	Purpose	Notes
Stand-alone Primary Site	Manages clients directly. It is the only site type required in many organizations.	Microsoft recommends this design wherever possible. Supports up to about 250,000 clients.

Site type	Purpose	Notes
Central Administration Site (CAS)	Sits at the top of a hierarchy for central reporting and administration. It does not manage clients itself.	Only needed beyond the limits of one primary site, with up to 25 child primary sites. Cannot be added later to an existing stand-alone site.
Secondary Site	A small site below a primary site for remote locations with limited bandwidth.	Often replaced today by a simple Distribution Point with boundary groups.

Role-Based Administration

SCCM controls who can do what using **Role-Based Administration (RBAC)**. Administrators are given **security roles** (what actions they may perform), **security scopes** (which objects they can see, such as applications or boot images) and **collections** (which devices or users they can act on). Typical built-in roles include Full Administrator, Application Administrator, Operations Administrator, Asset Manager and Read-only Analyst. For example, a service desk member can be allowed to deploy software to one branch's computers only, without any access to the rest of the environment.

SCCM Site Systems

A **site system** is any server that provides services to a site, and each service it provides is called a **site system role**. Roles can be combined on one server or distributed according to the size of the environment.

Site system role	Function
Management Point	Handles all client communication: policy requests, inventory, status messages and location of content.
Distribution Point	Stores and serves content. Supports content validation, bandwidth control and PXE for operating system deployment.
Software Update Point	Works with WSUS to synchronize update metadata so SCCM can scan clients and deploy updates.
Reporting Services Point	Integrates SQL Server Reporting Services (SSRS) so reports can be viewed from the console and a web portal.
State Migration Point	Temporarily stores user state data during an operating system replacement or refresh.
Fallback Status Point	Receives client status messages when a client cannot reach its Management Point, which helps with troubleshooting installation problems.
Endpoint Protection Point	Enables Endpoint Protection management and the Microsoft Defender features in the console.

Site system role	Function
Cloud Management Gateway	Allows Internet-based clients (for example remote users not on VPN) to be managed through Azure.

For best performance, Distribution Points should be placed close to the clients that use them, such as in each branch, while the site server and database stay in the data centre. Boundary groups then decide which Distribution Point a client uses.

SCCM Client Installation and Management

A computer is only managed once the SCCM client is installed. Before starting, make sure that the computer is discovered (visible in **Assets and Compliance > Devices**), is inside a boundary group, can resolve the site server by DNS and that the firewall allows File and Printer Sharing and the SCCM ports. The common installation methods are:

Method	Description
Client Push	The site server connects to discovered computers and installs the client automatically using the Client Push Installation Account. Simple to use, so it is the most common method in labs.
Group Policy	A startup script or software installation policy runs `ccmsetup.exe` on domain computers. Useful where the firewall blocks client push.
Software Update-Based	The client is published as a “software update” through WSUS and installed via Windows Update.
Manual	Run `ccmsetup.exe` locally from the `Client` folder on the site server share.
Operating System Imaging	The client is installed as a step in a task sequence during OS deployment.

An example of a manual installation command is shown below. It points the client to the Management Point and the site code:

```
ccmsetup.exe /mp:SCCM01.contoso.local SMSSITECODE=PS1
```

After installation the **Configuration Manager** applet appears in Control Panel on the client. The installation progress is recorded in `C:\Windows\ccmsetup\Logs\ccmsetup.log`, and the client’s approval and activity can be checked in the console, where the device should show **Client = Yes** and **Client Activity = Active**.

Managing devices

- **Collections:** devices and users are grouped into collections, either by direct membership or by a query (for example, “all computers whose name begins with PC-HR”, or “all computers running Windows 11”). Deployments and client settings are always targeted at collections.

- **Client settings:** control client behavior such as policy polling interval (60 minutes by default), hardware and software inventory, software updates and remote tools.
- **Deployments:** applications, software updates, packages, task sequences and configuration baselines are all deployed to collections as either **Available** (user chooses) or **Required** (installs automatically at the deadline).
- **Client tools:** from the console right-click menu administrators can trigger client notification actions, such as Machine Policy refresh, Resource Explorer, Remote Control and Run Script.

Reporting and Monitoring

Deploying software is only half of the job, because the administrator must also confirm that it actually worked. SCCM provides built-in tools for this under the **Monitoring** workspace:

- **System Status:** shows the health of the site and each component, with Site Status, Component Status and Status Message Queries to dig into errors.
- **Client Status:** shows how many clients are active or inactive and the results of the daily client health checks.
- **Deployments:** shows success, in-progress, error and unknown counts for every deployment, with details per device.
- **Distribution Status:** shows whether content has reached every Distribution Point.
- **Alerts:** triggered by conditions such as a failed site component or a low disk space. Email subscriptions can notify the team automatically.

For detailed reporting, the **Reporting Services Point** publishes hundreds of built-in reports covering hardware, software, software updates compliance, deployments, operating system and more. Reports can be exported to Excel or PDF, subscribed to by email on a schedule, or customized with Report Builder.

SCCM Logs and Troubleshooting

SCCM records almost everything it does in text log files. Learning which log relates to which problem is the most important troubleshooting skill for an SCCM administrator. Logs are read with **CMTrace** (or Support Center), which highlights errors in red and warnings in yellow, and updates in real time. On the site server logs are found in `<InstallDir>\Logs`, and on clients in `C:\Windows\CCM\Logs`.

Area	Log file	What it shows
Site installation	`ConfigMgrSetup.log`	Installation progress and failures during setup (C:\ root).
Client installation (server)	`ccm.log`	Client push attempts made by the site server.
Client installation (client)	`ccmsetup.log`	Client installer activity on the computer.
Policy	`PolicyAgent.log`, `PolicyEvaluator.log`	Downloading and evaluating policy on the client.

Area	Log file	What it shows
Site assignment and location	`ClientIDManagerStartup.log`, `LocationServices.log`	Client registration, site assignment and finding MPs/DPs.
Software deployment	`AppEnforce.log`, `AppDiscovery.log`, `execmgr.log`	Application detection, installation and execution results.
Content distribution	`distmgr.log`, `PkgXferMgr.log`	Content processing and transfer to Distribution Points (site server).
Software updates	`wsyncmgr.log`, `WUAHandler.log`, `UpdatesDeployment.log`	Update synchronization, scanning and installation.
Inventory	`InventoryAgent.log`	Hardware and software inventory on the client.
OS deployment	`smsts.log`	Task sequence steps during OS deployment.

Common problems and first checks

Problem	What to check first
Client does not install	Firewall and admin share access, the Client Push account, DNS resolution, and `ccm.log` / `ccmsetup.log`.
Client installed but inactive	CcmExec service running, Management Point reachable, and `ClientIDManagerStartup.log` and `LocationServices.log` for site assignment or boundary issues.
Application will not install	`AppDiscovery.log` and `AppEnforce.log` for detection method or command-line errors, and whether the content is available on a DP in the client's boundary group.
Content stuck in "In progress"	`distmgr.log` and `PkgXferMgr.log`, DP disk space and the DP's health in Monitoring.
Software updates not applying	Software Update Point synchronization (`wsyncmgr.log`), WSUS health, and `WUAHandler.log` on the client.
Site component errors	Monitoring > Component Status, `sitecomp.log` and free disk space or SQL health on the site server.

A good troubleshooting habit is to follow the data from beginning to end: the console, then the site server logs, then the Management Point or Distribution Point, and finally the client logs.

Testing every change first on a small pilot collection also prevents most large-scale deployment problems.